Server Tasks

1. **Install nginx and run nginx on port number 81.**

   Install nginx package

```
Warning: Permanently added '3.25.236.144' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-41-233 ~]$ sudo yum update
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-41-233 ~]$ sudo yum install nginx -y
Last metadata expiration check: 0:00:40 ago on Wed Oct 14 11:00:40 2026.
Dependencies resolved.
================================================================================
 Package                 Architecture            Version
================================================================================
Installing:
 nginx                   x86_64                  1:1.30.5-1.amzn2023.0.1
Installing dependencies:
 generic-logos-httpd     noarch                  18.0.0-12.amzn2023.0.3
 gperftools-libs         x86_64                  2.9.1-1.amzn2023.0.3
 libunwind               x86_64                  1.4.0-5.amzn2023.0.3
 nginx-core              x86_64                  1:1.30.5-1.amzn2023.0.1
 nginx-filesystem        noarch                  1:1.30.5-1.amzn2023.0.1
 nginx-mimetypes         noarch                  2.1.49-3.amzn2023.0.3

Transaction Summary
```

   Start the nginx package and check its status

```
ec2-user@ip-172-31-41-233:~
[ec2-user@ip-172-31-41-233 ~]$ sudo systemctl start nginx
[ec2-user@ip-172-31-41-233 ~]$ sudo systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
     Active: active (running) since Wed 2026-10-14 11:02:12 UTC; 11s ago
    Process: 25925 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
    Process: 25928 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
    Process: 25948 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
   Main PID: 25979 (nginx)
      Tasks: 3 (limit: 1059)
     Memory: 3.4M
        CPU: 82ms
     CGroup: /system.slice/nginx.service
             ├─25979 "nginx: master process /usr/sbin/nginx"
             ├─25983 "nginx: worker process"
             └─25984 "nginx: worker process"

Oct 14 11:02:12 ip-172-31-41-233.ap-southeast-2.compute.internal systemd[1]: Starting nginx.servic
Oct 14 11:02:12 ip-172-31-41-233.ap-southeast-2.compute.internal nginx[25928]: nginx: the configur
Oct 14 11:02:12 ip-172-31-41-233.ap-southeast-2.compute.internal nginx[25928]: nginx: configuratio
Oct 14 11:02:12 ip-172-31-41-233.ap-southeast-2.compute.internal systemd[1]: Started nginx.service
```

   Now configure the port in nginx.conf

   vi /etc/nginx/nginx.conf

```
http {
    log_format  main  '$remote_addr - $remote_user [$time_local] "$request" '
                      '$status $body_bytes_sent "$http_referer" '
                      '"$http_user_agent" "$http_x_forwarded_for"';

    access_log  /var/log/nginx/access.log  main;

    sendfile            on;
    tcp_nopush          on;
    keepalive_timeout   65;
    types_hash_max_size 4096;

    include             /etc/nginx/mime.types;
    default_type        application/octet-stream;

    # Load modular configuration files from the /etc/nginx/conf.d directory.
    # See http://nginx.org/en/docs/ngx_core_module.html#include
    # for more information.
    include /etc/nginx/conf.d/*.conf;

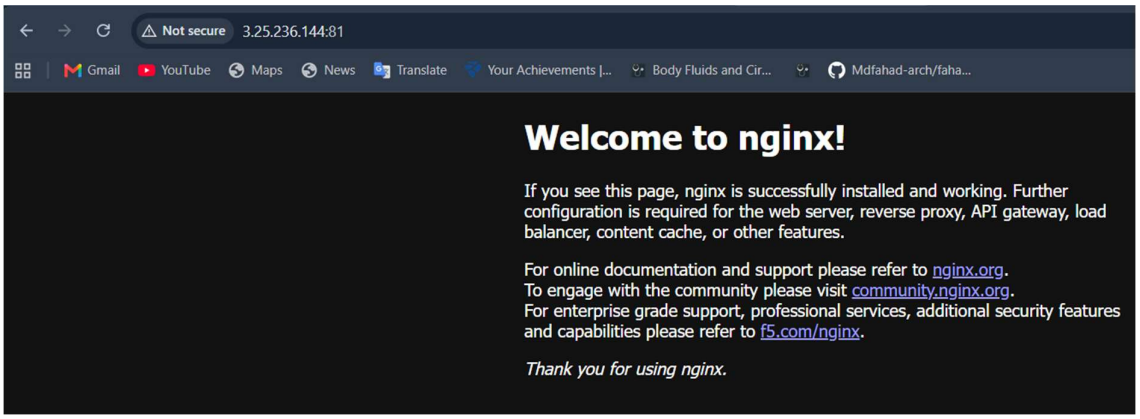
    server {
        listen       81;
        listen       [::]:81;
        server_name  _;
        root         /usr/share/nginx/html;

        # Load configuration files for the default server block.
        include /etc/nginx/default.d/*.conf;
```

Restart the package with systemctl

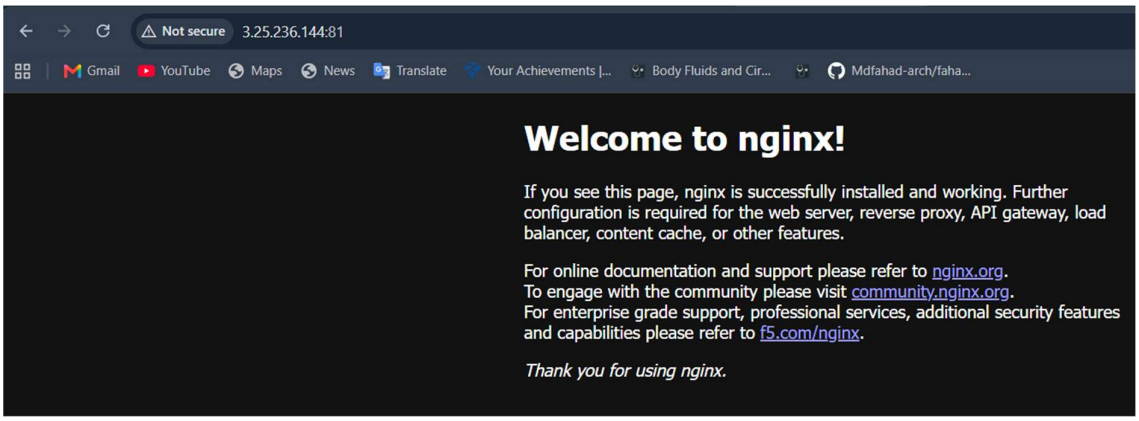
```
ec2-user@ip-172-31-41-233:/
[ec2-user@ip-172-31-41-233 /]$ sudo systemctl restart nginx
[ec2-user@ip-172-31-41-233 /]$
```

Open the port in web browser public Ip:80



2. Deploy a sample index.html file on nginx.

After starting the service open the web browser

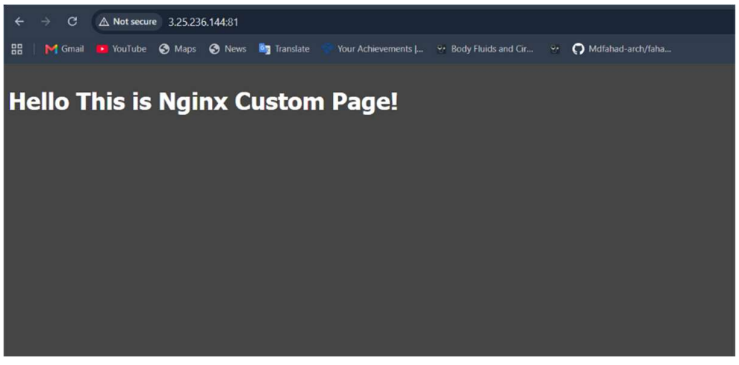


Now navigate to directory /usr/share/html

Then change the html in index.html at /usr/share/html/index.html

```
ec2-user@ip-172-31-41-233:/usr/share/nginx
[ec2-user@ip-172-31-41-233 nginx]$ sudo nano /usr/share/nginx/html/index.html
[ec2-user@ip-172-31-41-233 nginx]$
```

Refresh the web browser to see the new html page



3. Install Apache and run Apache on port number 82

Install Apache (httpd)

```
[ec2-user@ip-172-31-41-233 nginx]$ sudo yum install httpd -y
Last metadata expiration check: 1:20:08 ago on Wed Oct 14 11:00:40 2026.
Dependencies resolved.
================================================================================
 Package                 Architecture            Version
================================================================================
Installing:
 httpd                   x86_64                  2.4.69-1.amzn2023.0.2
Installing dependencies:
 apr                     x86_64                  1.7.5-1.amzn2023.0.4
 apr-util                x86_64                  1.6.5-1.amzn2023.0.1
 apr-util-lmdb           x86_64                  1.6.5-1.amzn2023.0.1
 httpd-core              x86_64                  2.4.69-1.amzn2023.0.2
 httpd-filesystem        noarch                  2.4.69-1.amzn2023.0.2
 httpd-tools             x86_64                  2.4.69-1.amzn2023.0.2
 libbrotli               x86_64                  1.1.0-1.amzn2023.0.1
 mailcap                 noarch                  2.1.49-3.amzn2023.0.3
Installing weak dependencies:
 apr-util-openssl        x86_64                  1.6.5-1.amzn2023.0.1
 mod_http2               x86_64                  1:2.4.69-1.amzn2023.0.2
 mod_lua                 x86_64                  2.4.69-1.amzn2023.0.2

Transaction Summary
================================================================================
Install  12 Packages
```

Starting the server package, checking the status

```
ec2-user@ip-172-31-41-233:/
[ec2-user@ip-172-31-41-233 /]$ sudo systemctl start httpd
[ec2-user@ip-172-31-41-233 /]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
     Active: active (running) since Wed 2026-10-14 12:22:21 UTC; 4min 4s ago
       Docs: man:httpd.service(8)
   Main PID: 29190 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 177 (limit: 1059)
     Memory: 13.8M
        CPU: 330ms
     CGroup: /system.slice/httpd.service
             ├─29190 /usr/sbin/httpd -DFOREGROUND
             ├─29191 /usr/sbin/httpd -DFOREGROUND
             ├─29192 /usr/sbin/httpd -DFOREGROUND
             ├─29193 /usr/sbin/httpd -DFOREGROUND
             └─29275 /usr/sbin/httpd -DFOREGROUND

Oct 14 12:22:21 ip-172-31-41-233.ap-southeast-2.compute.internal systemd[1]: Starting httpd.service - The Apache H
Oct 14 12:22:21 ip-172-31-41-233.ap-southeast-2.compute.internal systemd[1]: Started httpd.service - The Apache HT
Oct 14 12:22:21 ip-172-31-41-233.ap-southeast-2.compute.internal httpd[29190]: Server configured, listening on: po
[ec2-user@ip-172-31-41-233 /]$
```

To change the port of httpd server go to below path

```
ec2-user@ip-172-31-41-233:/etc/httpd
[ec2-user@ip-172-31-41-233 httpd]$ cat /etc/httpd/conf/httpd.conf
#
```

Then change the port from 80 to 82

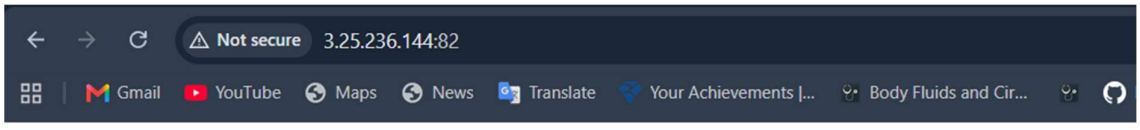
```
#
# Change this to Listen on a specific IP address, but note that if
# httpd.service is enabled to run at boot time, the address may not be
# available when the service starts.  See the httpd.service(8) man
# page for more information.
#
#Listen 12.34.56.78:80
Listen 82

#
```

But for this to work we have to restart the server

```
ec2-user@ip-172-31-41-233:/etc/httpd
[ec2-user@ip-172-31-41-233 httpd]$ sudo systemctl restart httpd
[ec2-user@ip-172-31-41-233 httpd]$
```

Then access it on webserver with port 82



It works!

4. **Deploy a sample index.html file on Apache.**

First start the server

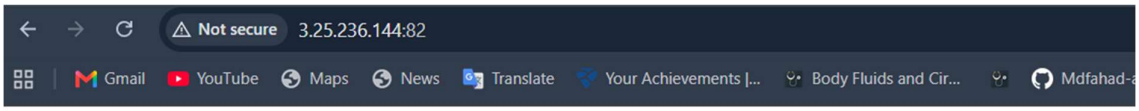
```
ec2-user@ip-172-31-41-233:/
[ec2-user@ip-172-31-41-233 /]$ sudo systemctl start httpd
[ec2-user@ip-172-31-41-233 /]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
     Active: active (running) since Wed 2026-10-14 12:22:21 UTC; 4min 4s ago
       Docs: man:httpd.service(8)
   Main PID: 29190 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 177 (limit: 1059)
     Memory: 13.8M
        CPU: 330ms
     CGroup: /system.slice/httpd.service
             ├─29190 /usr/sbin/httpd -DFOREGROUND
             ├─29191 /usr/sbin/httpd -DFOREGROUND
             ├─29192 /usr/sbin/httpd -DFOREGROUND
             ├─29193 /usr/sbin/httpd -DFOREGROUND
             └─29275 /usr/sbin/httpd -DFOREGROUND

Oct 14 12:22:21 ip-172-31-41-233.ap-southeast-2.compute.internal systemd[1]: Starting httpd.service - The Apache H
Oct 14 12:22:21 ip-172-31-41-233.ap-southeast-2.compute.internal systemd[1]: Started httpd.service - The Apache HT
Oct 14 12:22:21 ip-172-31-41-233.ap-southeast-2.compute.internal httpd[29190]: Server configured, listening on: po
[ec2-user@ip-172-31-41-233 /]$
```

To deploy the index.html check for path /var/www/html

```
ec2-user@ip-172-31-41-233:~
[ec2-user@ip-172-31-41-233 share]$ sudo nano /var/www/html/index.html
```

Make changes in the file and then refresh page to see changes



Hello this is custom page of apache2

5. **Install Apache tomcat on port number 8082**

Install the tomcat package with Yum

```
ec2-user@ip-172-31-41-233:~
[ec2-user@ip-172-31-41-233 ~]$ sudo dnf install tomcat10 -y
Last metadata expiration check: 2:19:19 ago on Wed Oct 14 11:00:40 2026.
Dependencies resolved.
================================================================================
 Package                                Architecture                Version
================================================================================
Installing:
 tomcat10                               noarch                      1:10.1.60-1.amzn2023.0.1
Installing dependencies:
 cairo                                  x86_64                      1.18.0-4.amzn2023.0.3
 ecj                                    noarch                      1:4.23-1.amzn2023.0.4
 fonts-filesystem                       noarch                      1:2.0.5-12.amzn2023.0.2
 google-noto-fonts-common               noarch                      20240401-1.amzn2023.0.2
 google-noto-sans-vf-fonts              noarch                      20240401-1.amzn2023.0.2
 graphite2                              x86_64                      1.3.14-7.amzn2023.0.3
 harfbuzz                               x86_64                      7.0.0-2.amzn2023.0.2
 java-17-amazon-corretto-headless       x86_64                      1:17.0.20+12-1.amzn2023.1
 javapackages-filesystem                noarch                      6.0.0-7.amzn2023.0.6
 javapackages-tools                     noarch                      6.0.0-7.amzn2023.0.6
 langpacks-core-font-en                 noarch                      3.0-21.amzn2023.0.4
 libX11                                 x86_64                      1.8.10-2.amzn2023.0.2
 libX11-common                          noarch                      1.8.10-2.amzn2023.0.2
 libXau                                 x86_64                      1.0.11-6.amzn2023.0.1
 libXext                                x86_64                      1.3.6-1.amzn2023.0.1
 libXrender                             x86_64                      0.9.11-6.amzn2023.0.2
 libjpeg-turbo                          x86_64                      2.1.4-2.amzn2023.0.5
 libpng                                 x86_64                      2:1.6.37-10.amzn2023.0.13
 libxcb                                 x86_64                      1.17.0-1.amzn2023.0.1
 pixman                                 x86_64                      0.43.4-1.amzn2023.0.4
 tomcat10-el-5.0-api                    noarch                      1:10.1.60-1.amzn2023.0.1
 tomcat10-jsp-3.1-api                   noarch                      1:10.1.60-1.amzn2023.0.1
 tomcat10-lib                           noarch                      1:10.1.60-1.amzn2023.0.1
 tomcat10-servlet-6.0-api               noarch                      1:10.1.60-1.amzn2023.0.1
 xml-common                             noarch                      0.6.3-56.amzn2023.0.2
Installing weak dependencies:
 alsa-lib                               x86_64                      1.2.7.2-1.amzn2023.0.4
 dejavu-sans-fonts                      noarch                      2.37-16.amzn2023.0.2
 dejavu-sans-mono-fonts                 noarch                      2.37-16.amzn2023.0.2
 dejavu-serif-fonts                     noarch                      2.37-16.amzn2023.0.2
 fontconfig                             x86_64                      2.13.94-2.amzn2023.0.2
 freetype                               x86_64                      2.13.2-5.amzn2023.0.2
 tomcat-native                          x86_64                      2.0.16-1.amzn2023.0.1

Transaction Summary
================================================================================
Install  33 Packages
```

Start the package and check its status with systemctl command

```
[root@ip-172-31-41-233 bin]# systemctl start tomcat.service
[root@ip-172-31-41-233 bin]# systemctl status tomcat.service
● tomcat.service - Apache Tomcat 10
     Loaded: loaded (/etc/systemd/system/tomcat.service; disabled; preset: disabled)
     Active: active (running) since Thu 2026-10-15 11:21:45 UTC; 12s ago
    Process: 5595 ExecStart=/opt/apache-tomcat-10.1.61/bin/startup.sh (code=exited, status=0/SUCCESS)
   Main PID: 5608 (java)
      Tasks: 29 (limit: 1059)
     Memory: 78.0M
        CPU: 4.548s
     CGroup: /system.slice/tomcat.service
             └─5608 /usr/lib/jvm/java-21-amazon-corretto/bin/java -Djava.util.logging.config.file=/opt/apache-tomcat-10.1.61/

Oct 15 11:21:45 ip-172-31-41-233.ap-southeast-2.compute.internal systemd[1]: Starting tomcat.service - Apache Tomcat 10...
Oct 15 11:21:45 ip-172-31-41-233.ap-southeast-2.compute.internal startup.sh[5595]: Tomcat started.
Oct 15 11:21:45 ip-172-31-41-233.ap-southeast-2.compute.internal systemd[1]: Started tomcat.service - Apache Tomcat 10.
lines 1-14/14 (END)
```

By default the port for tomcat is 8080. To change this we should change the port in conifg file of tomcat

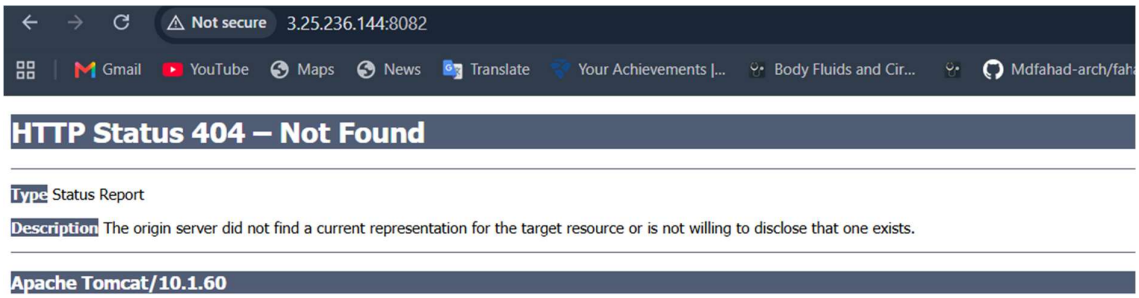
The path of this file is /etc/tomcat10/service/xml

```
[ec2-user@ip-172-31-41-233 tomcat10]$ sudo nano server.xml
[ec2-user@ip-172-31-41-233 tomcat10]$
```

Open the file and change the port in the http protocol

```
<!-- A "Connector" represents an endpoint by which requests are received
     and responses are returned. Documentation at :
     HTTP Connector: /docs/config/http.html
     AJP  Connector: /docs/config/ajp.html
     Define a non-SSL/TLS HTTP/1.1 Connector on port 8080
-->
<Connector port="8082" protocol="HTTP/1.1"
           connectionTimeout="20000"
           redirectPort="8443"
           maxParameterCount="1000"
```

After changing the port we have to restart the service with systemctl for it work



6. Deploy a sample app on webapps

To deploy our sample app we have to find webapps folder in the tomcat

```
root@ip-172-31-41-233:/opt/apache-tomcat-10.1.61/webapps
[root@ip-172-31-41-233 ~]# cd /opt/apache-tomcat-10.1.61/webapps
```

Create a folder in that directory and extract the sample.war or the custom app in that folder.

```
[root@ip-172-31-41-233 webapps]# mkdir sample
[root@ip-172-31-41-233 webapps]# ls
ROOT  docs  examples  host-manager  manager  sample
[root@ip-172-31-41-233 webapps]# cd sample
[root@ip-172-31-41-233 sample]# wget https://tomcat.apache.org/tomcat-6.0-doc/appdev/s
--2026-10-15 10:49:45--  https://tomcat.apache.org/tomcat-6.0-doc/appdev/sample/sample
Resolving tomcat.apache.org (tomcat.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to tomcat.apache.org (tomcat.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 4606 (4.5K)
Saving to: 'sample.war'

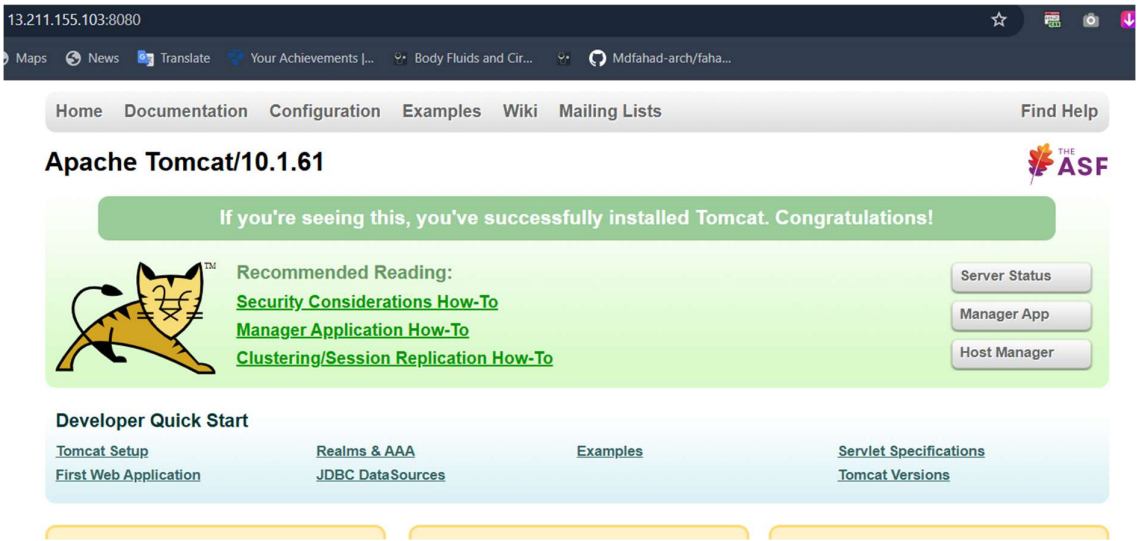
sample.war                                          100%[=========================

2026-10-15 10:49:46 (45.2 MB/s) - 'sample.war' saved [4606/4606]
```

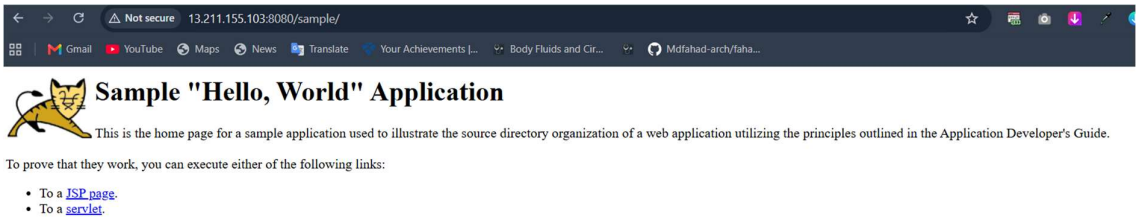
For compressed files other than zip or tar we use unzip command

```
[root@ip-172-31-41-233 sample]# unzip sample.war
Archive:  sample.war
   creating: META-INF/
  inflating: META-INF/MANIFEST.MF
   creating: WEB-INF/
   creating: WEB-INF/classes/
   creating: WEB-INF/classes/mypackage/
   creating: WEB-INF/lib/
   creating: images/
  inflating: WEB-INF/classes/mypackage/Hello.class
  inflating: WEB-INF/web.xml
  inflating: hello.jsp
  inflating: images/tomcat.gif
  inflating: index.html
[root@ip-172-31-41-233 sample]# ls
META-INF  WEB-INF  hello.jsp  images  index.html  sample.war
```

Now when we open the webpage after starting the service. We see



If we navigate or change the path in the web-url we can see our webapps



7. Create a tomcat.service file for tomcat.

Find the service file in the /etc/systemd/system/ path if there is no service file create a tomcat.service

```
root@ip-172-31-41-233:/opt/apache-tomcat-10.1.61/bin
[root@ip-172-31-41-233 bin]# sudo vi /etc/systemd/system/tomcat.service
[root@ip-172-31-41-233 bin]# cat /etc/systemd/system/tomcat.service

[Unit]
Description=Apache Tomcat 10
After=network.target

[Service]
Type=forking

Environment="JAVA_HOME=/usr/lib/jvm/java-21-amazon-corretto"
Environment="CATALINA_HOME=/opt/apache-tomcat-10.1.61"
Environment="CATALINA_BASE=/opt/apache-tomcat-10.1.61"

ExecStart=/opt/apache-tomcat-10.1.61/bin/startup.sh
ExecStop=/opt/apache-tomcat-10.1.61/bin/shutdown.sh

Restart=on-failure

[Install]
WantedBy=multi-user.target

[root@ip-172-31-41-233 bin]#
```

Then use daemon-reload to configure the service file

```
[ec2-user@ip-172-31-41-233 ~]$ sudo systemctl daemon-reload
```

After this try to start tomcat.service file or tomcat using systemctl

```
[root@ip-172-31-41-233 bin]# systemctl start tomcat.service
[root@ip-172-31-41-233 bin]# systemctl status tomcat.service
● tomcat.service - Apache Tomcat 10
     Loaded: loaded (/etc/systemd/system/tomcat.service; disabled; preset: disabled)
     Active: active (running) since Thu 2026-10-15 11:21:45 UTC; 12s ago
    Process: 5595 ExecStart=/opt/apache-tomcat-10.1.61/bin/startup.sh (code=exited, status=0/SUCCESS)
   Main PID: 5608 (java)
      Tasks: 29 (limit: 1059)
     Memory: 78.0M
        CPU: 4.548s
     CGroup: /system.slice/tomcat.service
             └─5608 /usr/lib/jvm/java-21-amazon-corretto/bin/java -Djava.util.logging.config.file=/opt/apache-tomcat-10.1.61/

Oct 15 11:21:45 ip-172-31-41-233.ap-southeast-2.compute.internal systemd[1]: Starting tomcat.service - Apache Tomcat 10...
Oct 15 11:21:45 ip-172-31-41-233.ap-southeast-2.compute.internal startup.sh[5595]: Tomcat started.
Oct 15 11:21:45 ip-172-31-41-233.ap-southeast-2.compute.internal systemd[1]: Started tomcat.service - Apache Tomcat 10.
lines 1-14/14 (END)
```

8. Configure HA Proxy server

Create three instances named server-1, server-2 and haproxy server

Server-1

Install nginx package and start the package with systemctl

```
root@ip-172-31-46-106:~
[root@ip-172-31-46-106 ~]# sudo yum install nginx
Last metadata expiration check: 0:02:29 ago on Thu Oct 15 11:31:50 2026.
Dependencies resolved.
================================================================================
 Package                    Architecture                Version
================================================================================
Installing:
 nginx                      x86_64                      1:1.30.5-1.amzn2023.0.1
Installing dependencies:
 generic-logos-httpd        noarch                      18.0.0-12.amzn2023.0.3
 gperftools-libs            x86_64                      2.9.1-1.amzn2023.0.3
 libunwind                  x86_64                      1.4.0-5.amzn2023.0.3
 nginx-core                 x86_64                      1:1.30.5-1.amzn2023.0.1
 nginx-filesystem           noarch                      1:1.30.5-1.amzn2023.0.1
 nginx-mimetypes            noarch                      2.1.49-3.amzn2023.0.3

Transaction Summary
================================================================================
Install  7 Packages

Total download size: 1.1 M
Installed size: 3.8 M
```

After starting the server add the public server of the haproxy instance

```
root@ip-172-31-46-106:~
[root@ip-172-31-46-106 ~]# vi /etc/hosts
[root@ip-172-31-46-106 ~]# cat /etc/hosts
127.0.0.1   localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost6 localhost6.localdomain6
3.25.89.50   loadbalancer
[root@ip-172-31-46-106 ~]#
```

Use ping command to see if loadbalancer or haproxy server is working or not

```
root@ip-172-31-46-106:~
[root@ip-172-31-46-106 ~]# ping loadbalancer -c4
PING loadbalancer (3.25.89.50) 56(84) bytes of data.

--- loadbalancer ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3147ms
```

After this go to server-2

Server-2

Now install httpd in this server and start the server with systemctl

```
[ec2-user@ip-172-31-35-226 ~]$ sudo yum update
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-35-226 ~]$ sudo yum install httpd -y
Last metadata expiration check: 0:00:29 ago on Thu Oct 15 11:44:41 2026.
Dependencies resolved.
================================================================================
 Package                 Architecture            Version
================================================================================
Installing:
 httpd                   x86_64                  2.4.69-1.amzn2023.0.2
Installing dependencies:
 apr                     x86_64                  1.7.5-1.amzn2023.0.4
 apr-util                x86_64                  1.6.5-1.amzn2023.0.1
 apr-util-lmdb           x86_64                  1.6.5-1.amzn2023.0.1
 generic-logos-httpd     noarch                  18.0.0-12.amzn2023.0.3
 httpd-core              x86_64                  2.4.69-1.amzn2023.0.2
 httpd-filesystem        noarch                  2.4.69-1.amzn2023.0.2
 httpd-tools             x86_64                  2.4.69-1.amzn2023.0.2
 libbrotli               x86_64                  1.1.0-1.amzn2023.0.1
 mailcap                 noarch                  2.1.49-3.amzn2023.0.3
Installing weak dependencies:
 apr-util-openssl        x86_64                  1.6.5-1.amzn2023.0.1
 mod_http2               x86_64                  1:2.4.69-1.amzn2023.0.2
 mod_lua                 x86_64                  2.4.69-1.amzn2023.0.2

Transaction Summary
================================================================================
Install  13 Packages
```

After starting the server add the haproxy server public ip into the /etc/hosts file

```
ec2-user@ip-172-31-35-226:~
[ec2-user@ip-172-31-35-226 ~]$ sudo vi /etc/hosts
[ec2-user@ip-172-31-35-226 ~]$ cat /etc/hosts
127.0.0.1   localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost6 localhost6.localdomain6
32.236.226.45 loadbalancer
[ec2-user@ip-172-31-35-226 ~]$
```

To check whether the load balancer or ha proxy server is working or not use ping to check the connectivity

```
[ec2-user@ip-172-31-35-226 ~]$ ping loadbalancer -c4
PING loadbalancer (32.236.226.45) 56(84) bytes of data.

--- loadbalancer ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3102ms

[ec2-user@ip-172-31-35-226 ~]$
```

Haproxy server

Now to set up the load balancer install haproxy package, and start the server with systemctl

```
[ec2-user@ip-172-31-35-2 ~]$ sudo yum update
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-35-2 ~]$ sudo yum install haproxy
Last metadata expiration check: 0:02:24 ago on Thu Oct 15 12:25:56 2026.
Dependencies resolved.
================================================================================
 Package                                Architecture
================================================================================
Installing:
 haproxy                                x86_64

Transaction Summary
================================================================================
Install  1 Package

Total download size: 2.7 M
Installed size: 8.5 M
Is this ok [y/N]: y
Downloading Packages:
haproxy-3.0.23-2.amzn2023.0.1.x86_64.rpm
--------------------------------------------------------------------------------
```

Now add the public Ip of both server-1 and server-2 into the /etc/hosts

```
[ec2-user@ip-172-31-35-2 ~]$ cat /etc/hosts
127.0.0.1   localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost6 localhost6.localdomain6
3.25.89.50 server-1
3.26.182.31 server-2
[ec2-user@ip-172-31-35-2 ~]$
```

Change the port bind from 5000 to 80 and also add both IP address of server-1 and server-2 into the backend apps with port 80

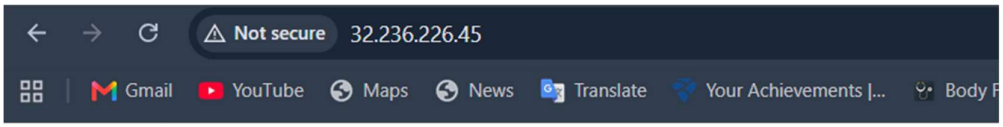
```
#---------------------------------------------------------------------
# main frontend which proxys to the backends
#---------------------------------------------------------------------
frontend main
    bind *:80
    acl url_static       path_beg       -i /static /images /javascript /stylesheets
    acl url_static       path_end       -i .jpg .gif .png .css .js

    use_backend static          if url_static
    default_backend             app

#---------------------------------------------------------------------
# static backend for serving up images, stylesheets and such
#---------------------------------------------------------------------
backend static
    balance     roundrobin
    server      static 127.0.0.1:4331 check

#---------------------------------------------------------------------
# round robin balancing between the various backends
#---------------------------------------------------------------------
backend app
    balance     roundrobin
    server  app1 127.0.0.1:5001 check
    server  app2 127.0.0.1:5002 check
    server  app3 127.0.0.1:5003 check
    server  app4 127.0.0.1:5004 check
     server app5 3.25.89.50:80 check
      server app6 3.26.182.31:80 check
[ec2-user@ip-172-31-35-2 ~]$
```

Now if we open haproxy server ip address in web browser we can see one server page after reloading it multiple times we can see other server



It works!

